Definition 2.6

Let r and θ be polar coordinates of the point $P(x, y)$ that corresponds to a non-zero complex number $z = x + iy$. The polar form or trigonometric form of a complex number P is

$$z = r(\cos\theta + i\sin\theta).$$

For convenience, we can write polar form as

$$z = x + iy = r(\cos\theta + i\sin\theta) = r\,cis\,\theta.$$

The value r represents the absolute value or modulus of the complex number z. The angle θ is called the argument or amplitude of the complex number z denoted by $\theta = \arg(z)$.

(i) If $z = 0$, the argument θ is undefined; and so it is understood that $z \neq 0$ whenever polar coordinates are used.

(ii) If the complex number $z = x + iy$ has polar coordinates (r, θ), its conjugate $\bar{z} = x - iy$ has polar coordinates $(r, -\theta)$.

Squaring and adding (1) and (2), and taking square root, the value of r is given by $r = |z| = \sqrt{x^2 + y^2}$.

Dividing (2) by (1), $\dfrac{r\sin\theta}{r\cos\theta} = \dfrac{y}{x} \Rightarrow \tan\theta = \dfrac{y}{x}$.

Case (i) The real number θ represents the angle, measured in radians, that z makes with the positive real axis when z is interpreted as a radius vector. The angle θ has an infinitely many possible values, including negative ones that differ by integral multiples of 2π. Those values can be determined from the equation $\tan\theta = \dfrac{y}{x}$ where the quadrant containing the point corresponding to z must be specified. Each value of θ is called an *argument* of z, and the set of all such values is obtained by adding multiple of 2π to θ, and it is denoted by arg z.

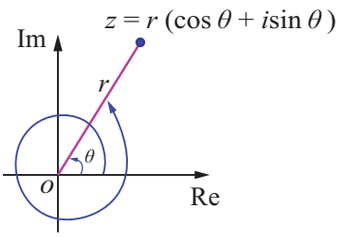


Fig. 2.29

Case (ii) There is a unique value of θ which satisfies the condition $-\pi < \theta \leq \pi$.

This value is called a principal value of θ or principal argument of z and is denoted by Arg z.

Note that,

$$-\pi < \text{Arg}(z) \leq \pi \quad or \quad -\pi < \theta \leq \pi$$

Principal Argument of a complex number

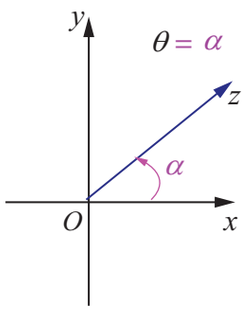
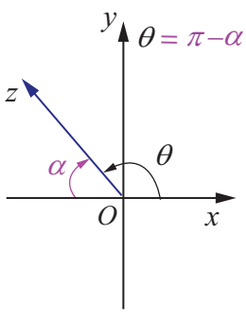
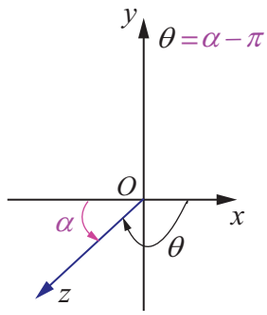
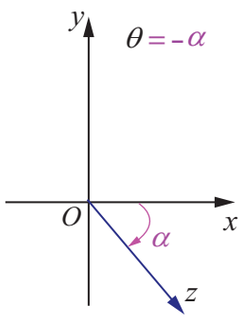
I-Quadrant	II-Quadrant	III-Quadrant	IV-Quadrant
			
$\theta = \alpha$	$\theta = \pi - \alpha$	$\theta = \alpha - \pi$	$\theta = -\alpha$

Fig. 2.30

Fig. 2.31

Fig. 2.32

Fig. 2.33

Chapter 2 Complex Numbers.indd 76 31-01-2020 17:33:54